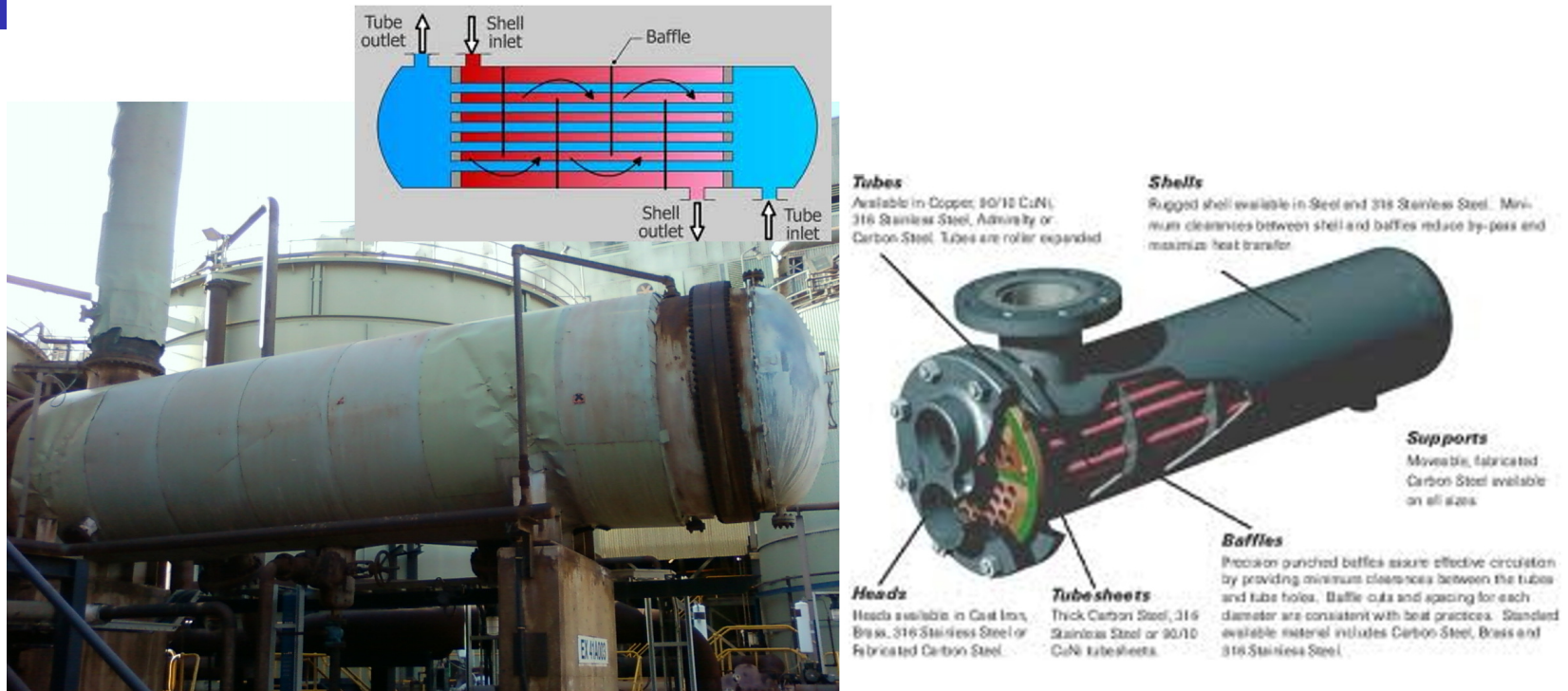


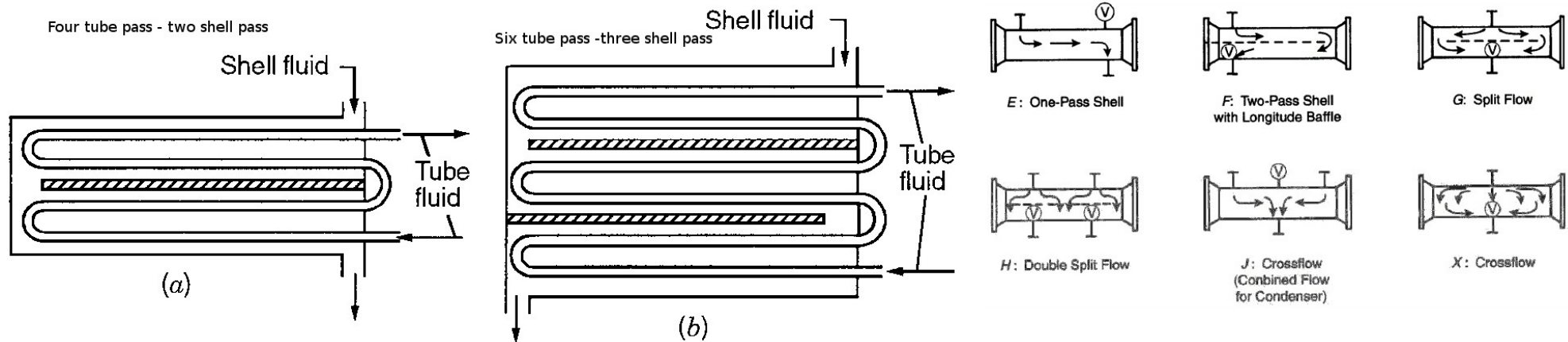
Shell and Tube Heat Exchangers



Shell and tube

- One fluid on the tube side and other fluid on the shell side
- Widely used type of heat exchangers in the industry
- Easy to clean, repair and maintenance
- International design codes TEMA, ASME

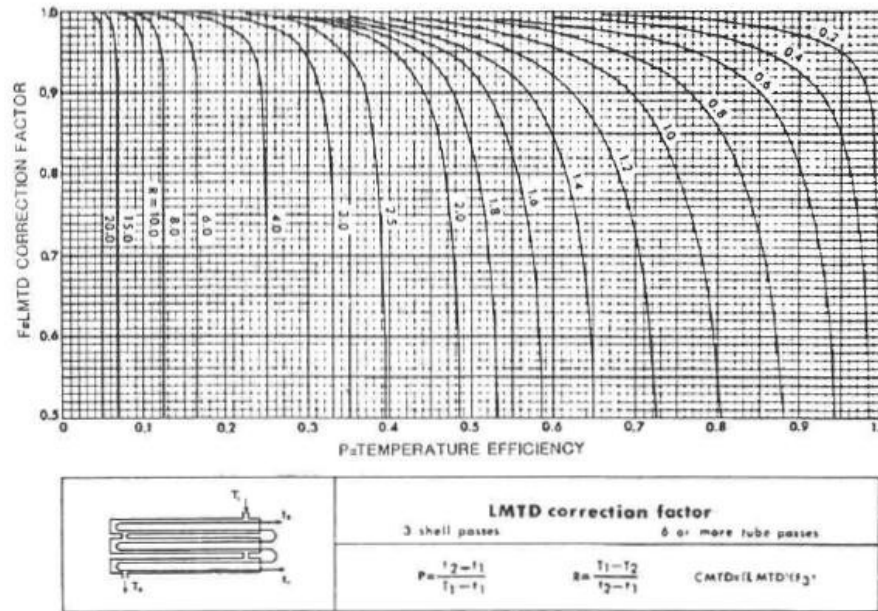
Shell and Tube Heat Exchangers Design



Design

- Assume the heat exchanger is counter flow heat exchanger
- Find the LMTD ΔT_m using Eq. (19)
- Find the correction factor from graph F corresponding to the type of shell and tube heat exchanger
- $Q = UAF\Delta T_m$

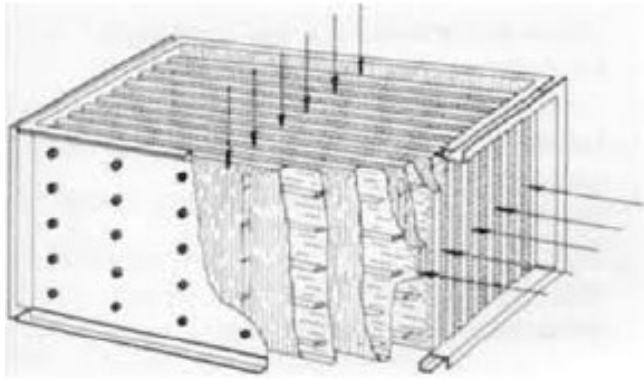
Shell and Tube Heat Exchangers Design



LMTD Correction

- Find the number of tube pass, shell pass
- Choose the graph
- Find $P = \frac{T_{c,out} - T_{c,in}}{T_{h,in} - T_{c,in}}$ and $R = \frac{T_{h,in} - T_{h,out}}{T_{c,out} - T_{c,in}}$
- Find the value F and Heat transfer $Q = UAF\Delta T_m$
- $F = 1$ for boiling or condensation

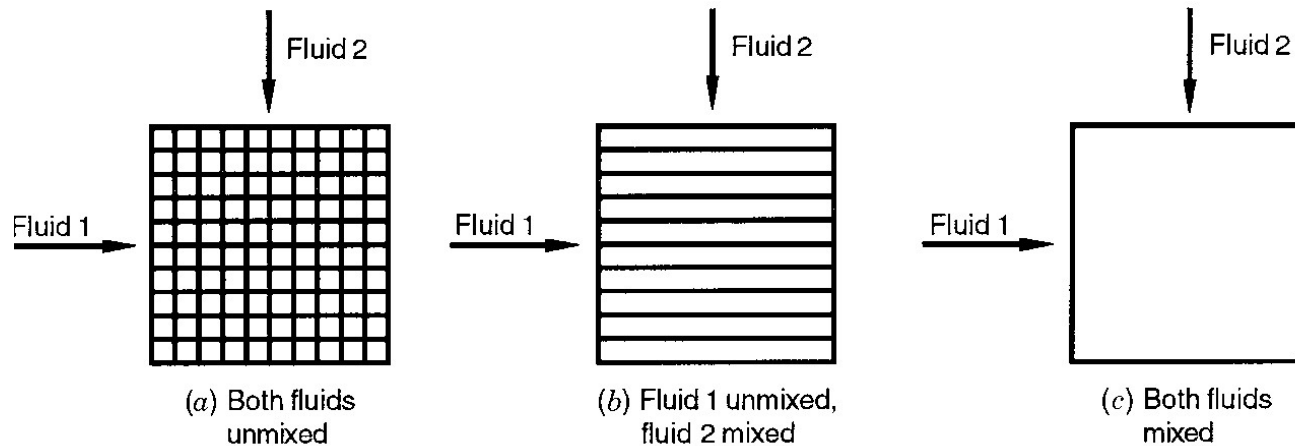
Cross Flow Heat Exchangers



Cross

- One fluid on one side and other fluid on the other side flow in a cross pattern
- Widely used type of heat exchangers in the industry, space, automobile, Air conditioners, etc.
- face difficulty to clean, repair and maintenance
- Mostly compact type of heat exchangers

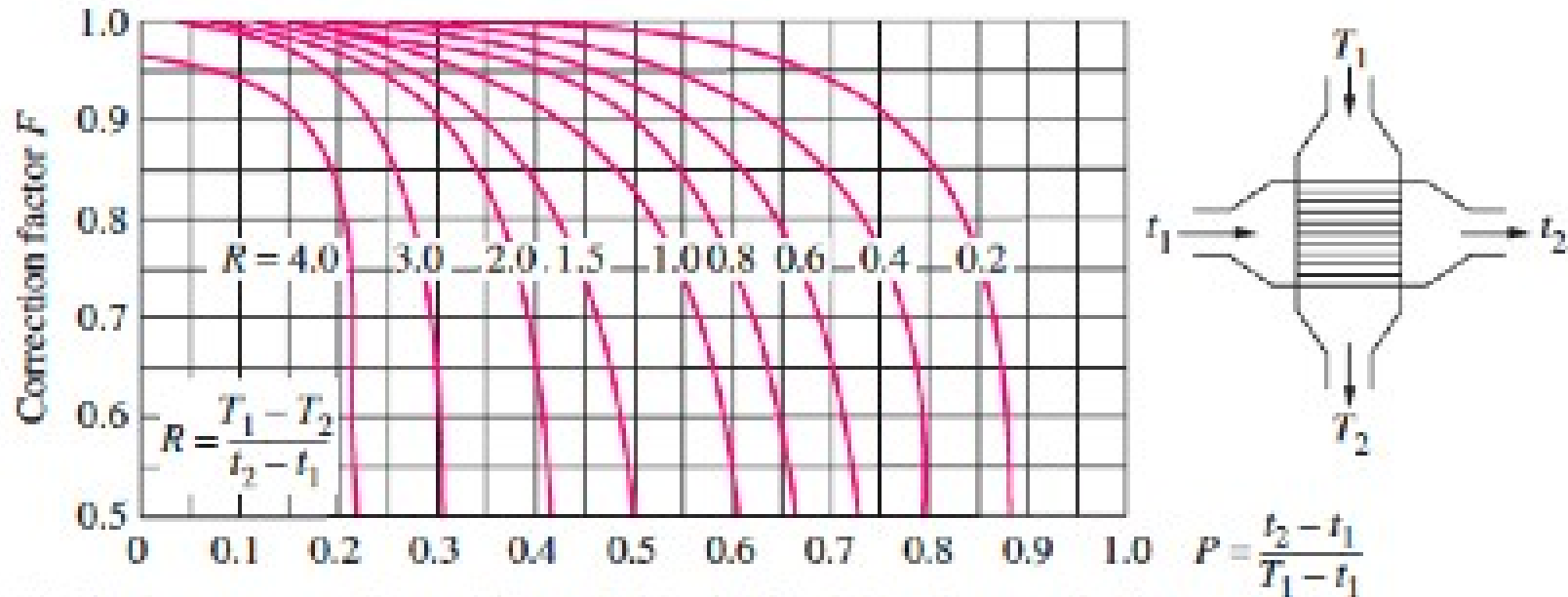
Cross Flow Heat Exchangers Design



Design

- Assume the heat exchanger is counter flow heat exchanger
- Find the LMTD ΔT_m using Eq. (19)
- Find the correction factor from graph F corresponding to the type of cross flow heat exchanger
- $Q = UAF\Delta T_m$, $F = 1$ for boiling or condensation

Cross Flow Heat Exchangers Design



(d) Single-pass cross-flow with one fluid *mixed* and the other *unmixed*

LMTD Correction

- Find the type - Fluid mixed or un-mixed
- Choose the graph
- Find $P = \frac{T_{c,out} - T_{c,in}}{T_{h,in} - T_{c,in}}$ and $R = \frac{T_{h,in} - T_{h,out}}{T_{c,out} - T_{c,in}}$
- Find the value F and Heat transfer $Q = UAF\Delta T_m$